PLANT DISEASE ANALYZER IN AGRICULTURE

23EC603-A PROTOTYPE LAB REPORT

Submitted by

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MARCH 2026

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Submitted for the Project viva-voce examination held on _____

INTERNAL EXAMINER

ABSTRACT

Plant diseases significantly affect agricultural productivity, making early detection and proper treatment very important. This project proposes a smart plant monitoring and disease detection system that combines both hardware and software approaches. The system is divided into two main parts: hardware implementation and software-based image processing. In the hardware section, different sensors are used to continuously monitor important environmental parameters such as plant temperature, humidity, soil moisture, and volatile gas levels. When the system starts, these parameters are displayed and monitored in real time. A threshold value of 200 is set for the gas sensor. If the volatile gas level exceeds this threshold, it indicates that the plant may be affected by a disease because plants often release certain gases when they are stressed or infected. Once the threshold is exceeded, the system automatically activates a pump to supply fertilizer or treatment to the plant. In the software section, a web-based application is developed for plant disease identification using image processing. The user can upload an image of the plant through the web interface, and the system analyzes the image to detect possible plant diseases. By integrating sensor-based monitoring with image-based disease detection, the proposed system provides an efficient and automated solution for early plant disease detection and management, helping to improve plant health and agricultural productivity.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL OVERVIEW

This project introduces an intelligent plant health monitoring and disease prediction system designed to support better agricultural management. The system combines sensor-based hardware monitoring with a web-based software platform to observe plant conditions and identify diseases at an early stage. The system is mainly focused on three types of crops: rice, banana, and sugarcane.

The project consists of two main sections: hardware implementation and software implementation. In the hardware section, different sensors are used to measure environmental parameters such as temperature, humidity, soil moisture, and volatile gas levels around the plant. These values are displayed when the system starts and are continuously monitored. A gas threshold value of 200 is predefined in the system. When the detected gas level goes beyond this limit, it may indicate that the plant is experiencing stress or disease conditions. In response, the system automatically activates a pump to supply fertilizer, helping to protect the plant and support its growth. The software section includes a web-based application that allows users to interact with the system easily. The website contains several pages including signup, login, home, and plant disease prediction pages. Users can register and log in to access the system features. Through the prediction page, users can upload images of rice, banana, or sugarcane plants, and the system processes these images to identify possible plant diseases using image analysis techniques. By integrating environmental monitoring through sensors and disease detection through image processing, the system provides a smart and efficient approach for monitoring plant conditions and predicting diseases in important crops.

1.2 PROBLEM DEFINITION

Agriculture plays a vital role in food production and the economy, but crop diseases significantly affect plant growth and reduce overall yield. One of the major challenges faced by farmers is the difficulty in identifying plant diseases at an early stage. Many plant diseases show symptoms on leaves, stems, or fruits, but farmers may not always have the knowledge or tools to accurately recognize these symptoms. As a result, diseases may spread quickly across the field, leading to severe crop damage and economic loss. In traditional farming methods, disease detection is usually done through manual observation or consultation with agricultural experts. This process can be time-consuming and may not always provide immediate results. Farmers in remote areas may also face difficulties in accessing expert guidance. Because of this delay, appropriate treatment or preventive measures may not be applied in time, which further increases the risk of crop loss. Another important factor affecting plant health is the surrounding environmental conditions. Parameters such as temperature, humidity, soil moisture, and the presence of harmful gases can influence plant growth and disease development. Without continuous monitoring of these parameters, farmers may not be aware of unfavorable conditions that can stress the plant or trigger disease symptoms. With the advancement of technology, it is possible to develop intelligent systems that combine sensor-based monitoring with artificial intelligence techniques. Sensors can continuously monitor environmental conditions and detect unusual changes, while image processing and machine learning models can analyze plant leaf images to identify diseases. Such a system can provide faster and more accurate detection compared to traditional methods. Therefore, there is a need to design a smart plant health monitoring and disease detection system that can help farmers easily identify plant diseases and monitor plant conditions in real time. By integrating hardware sensors with a web-based

AI platform, farmers can receive quick information about plant health and take timely action to protect their crops, ultimately improving agricultural productivity and reducing crop losses.

1.3 OBJECTIVES OF THE PROJECT

The main objectives of the Plant Disease Analyzer in Agriculture are:

1. To develop a smart plant monitoring system that can observe important environmental conditions affecting plant growth.

2. To measure key parameters such as temperature, humidity, soil moisture, and volatile gas levels using sensors.

3. To detect possible plant stress or disease by monitoring volatile gas levels and comparing them with a predefined threshold value.

4. To automate fertilizer supply by activating a pump when the gas level exceeds the set threshold, helping the plant recover at an early stage.

5. To design a web-based application that allows users to easily access the system through signup and login pages.

6. To develop an image-based plant disease prediction module where users can upload plant images for analysis.

7. To identify diseases in specific crops, particularly rice, banana, and sugarcane, using image processing techniques.

8. To provide an easy and efficient solution for farmers or users to monitor plant health and detect diseases early.

1.4 SCOPE OF THE PROJECT

The scope of this project is to develop a smart plant monitoring and disease detection system that helps farmers identify plant health conditions and possible diseases at an early stage. The system combines hardware sensors and a web-based software platform to monitor environmental parameters and analyze plant leaf images. The hardware part of the system monitors important plant conditions such as temperature, humidity, soil moisture, and volatile gas levels using sensors. These parameters help in understanding the environmental conditions affecting plant growth. When the gas sensor value exceeds the predefined threshold, the system detects a possible stress condition in the plant and automatically activates the fertilizer pump. The software part of the project focuses on plant disease detection using image processing and artificial intelligence. Farmers can upload images of plant leaves through a web-based interface. The system analyzes the image and predicts whether the plant is healthy or affected by a disease. The project mainly focuses on detecting diseases in three crops: rice, sugarcane, and banana. This system aims to support farmers by providing an easy-to-use platform for plant health monitoring and disease detection. It can be further expanded in the future to support more crops, additional sensors, and improved disease detection models.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW OF EXISTING SYSTEMS

In traditional agricultural practices, plant disease detection and crop monitoring are mainly carried out through manual observation by farmers or agricultural experts. Farmers inspect plant leaves and crop conditions visually to identify symptoms such as spots, discoloration, or pest damage. However, this method requires experience and continuous field monitoring, which can be time-consuming and may lead to delayed detection of plant diseases.

Some existing systems use basic environmental monitoring devices to measure parameters such as soil moisture, temperature, and humidity. While these systems help in understanding environmental conditions, they often do not include advanced techniques for detecting plant diseases from leaf images.

Other modern systems apply image processing and machine learning techniques for plant disease detection. These systems analyze images of plant leaves to identify disease patterns. However, many of these solutions focus only on image analysis and do not integrate environmental monitoring or real-time alert systems. Therefore, most existing systems either rely on manual inspection or provide limited monitoring capabilities. There is a need for an integrated system that combines environmental sensing, image processing, and IoT technology to enable accurate disease detection and real-time crop monitoring.

2.2 IEEE PAPERS REVIEWED

The review of these papers provides valuable insights into existing technologies used for plant disease detection and crop monitoring. It highlights the advantages and limitations of current methods such as image processing, machine learning, and deep learning techniques. This analysis helps in identifying research gaps and supports the development of an efficient plant disease analyzer system integrated

with IoT and image processing technologies.

2.2.1 Paper 1: Plant Disease Detection Using Image Processing Techniques

This research paper presents a system that uses image processing techniques to detect plant diseases from leaf images. The system captures leaf images and processes them through steps such as image preprocessing, segmentation, feature extraction, and classification. It helps in identifying disease symptoms such as leaf spots, discoloration, and abnormal patterns. The study highlights that automated disease detection can assist farmers in early identification and treatment of plant diseases. However, the accuracy of the system largely depends on image quality and proper feature extraction methods.

2.2.2 Paper 2: Plant Disease Identification Using Machine Learning Algorithms

This paper proposes a machine learning-based approach for identifying plant diseases from leaf images. The system extracts features such as color, texture, and shape from the images and uses classification algorithms like Support Vector Machine (SVM) to identify different diseases. The model improves detection efficiency and reduces manual effort in crop monitoring. However, the system requires a well-trained dataset and proper model training to achieve high accuracy.

2.2.3 Paper 3: Deep Learning-Based Plant Disease Detection System

This research focuses on using deep learning techniques for automatic detection of plant diseases. Convolutional Neural Networks (CNN) are used to analyze leaf images and classify different types of diseases. The system can automatically learn important features from the images without manual feature extraction. The study shows that deep learning models provide higher accuracy in disease detection. However, these models require large datasets and higher computational resources.

2.2.4 Paper 4: IoT-Based Smart Agriculture Monitoring System

This study presents an IoT-based system for monitoring environmental parameters affecting crop health. The system uses sensors to measure soil moisture, temperature, and humidity, and transmits the data to a cloud platform for monitoring and analysis. The system helps farmers monitor crop conditions remotely and take timely actions. However, network connectivity and sensor calibration can affect system reliability.

2.2.5 Paper 5: Automated Plant Disease Detection Using Image Analysis

This paper discusses the development of an automated system that detects plant diseases using digital image analysis. The system processes leaf images and identifies infected regions by analyzing color variations and texture patterns. The method helps in detecting diseases at an early stage and reduces crop loss. However, environmental factors such as lighting conditions and image noise may affect the accuracy of disease detection systems.

2.3 KEY FINDINGS AND CONTRIBUTIONS

2.3.1 Challenges in Existing Systems

- Many existing plant disease detection systems rely on manual observation by farmers, which is time-consuming and prone to errors.
- Some systems focus only on image-based disease detection without integrating environmental monitoring, limiting their effectiveness.
- Advanced deep learning models require large datasets and high computational resources, making real-time deployment difficult in rural areas.
- Variations in lighting, background, and leaf orientation can reduce the accuracy of automated detection systems.

2.3.2 Key Findings

- Automated image processing and machine learning methods significantly improve the speed and accuracy of plant disease detection compared to manual inspection.
- Deep learning models such as CNNs can automatically extract features from leaf images, providing higher detection accuracy.
- IoT-based monitoring systems enable real-time tracking of environmental parameters like soil moisture, temperature, and humidity, which affect plant health.
- Integration of sensors with cloud platforms allows remote monitoring and timely intervention, enhancing crop management.

2.3.3 Contributions

- Highlighted the need for integrated systems that combine leaf image analysis with environmental sensing for more effective plant health monitoring.
- Demonstrated that automated detection can reduce crop loss and improve agricultural productivity.
- Provided insights into practical challenges in deploying smart agriculture systems in rural or resource-limited areas.
- Established a foundation for developing an IoT-enabled Plant Disease Analyzer that supports early detection, real-time monitoring, and automated interventions.

2.4 SUMMARY

The review of existing literature highlights the growing role of image processing, machine learning, deep learning, and IoT technologies in plant disease detection and smart agriculture. Traditional methods, which rely on manual inspection, are time-consuming, error-prone, and cannot provide continuous monitoring. Existing automated systems improve disease detection but face challenges such

as dependency on high-quality datasets, computational resource requirements, and sensitivity to environmental factors like lighting and leaf orientation.

The studies also show that integrating environmental sensors with image analysis can enhance the accuracy and timeliness of disease detection. IoT-based systems enable real-time monitoring of soil moisture, temperature, humidity, and other factors that affect plant health, allowing farmers to take proactive measures to protect crops.

Overall, the literature emphasizes the need for a comprehensive, cost-effective, and real-time plant disease monitoring system that combines automated leaf image analysis, environmental sensing, and IoT connectivity to support sustainable agriculture and improve crop productivity.

CHAPTER 3

PROPOSED SOLUTION AND METHODOLOGY

3.1 INTRODUCTION

The proposed system is designed to monitor plant health and detect diseases using a combination of environmental sensors, image processing, and IoT technology. It measures important parameters such as soil moisture, temperature, and air quality to determine plant stress conditions. Leaf images are captured and analyzed to identify disease symptoms such as spots, discoloration, and abnormal patterns. The sensor data and analysis results are processed using an ESP32 microcontroller and transmitted to a cloud platform for real-time monitoring. The system also provides alerts and recommendations to farmers through a mobile application, enabling timely interventions to improve crop productivity and support smart agriculture practices.

3.2 BLOCK DIAGRAM

3.2.1 Hardware Block Diagram

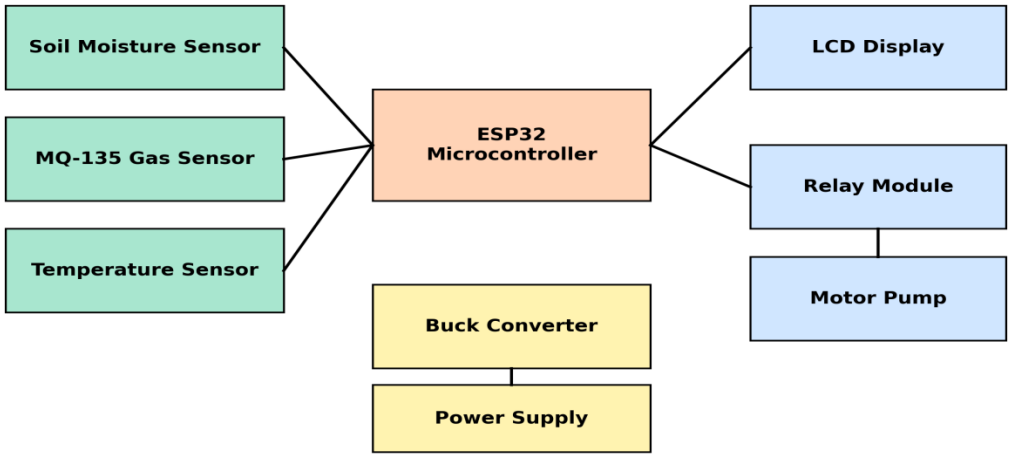


Fig 3.1 BLOCK DIAGRAM

The block diagram consists of the following components:

- Soil Moisture sensor
- MQ-135 sensor
- ESP32 Microcontroller
- Power Supply
- Buck Converter
- Temperature sensor
- Motor pump
- Relay module
- LCD display

3.2.2 Software Block Diagram

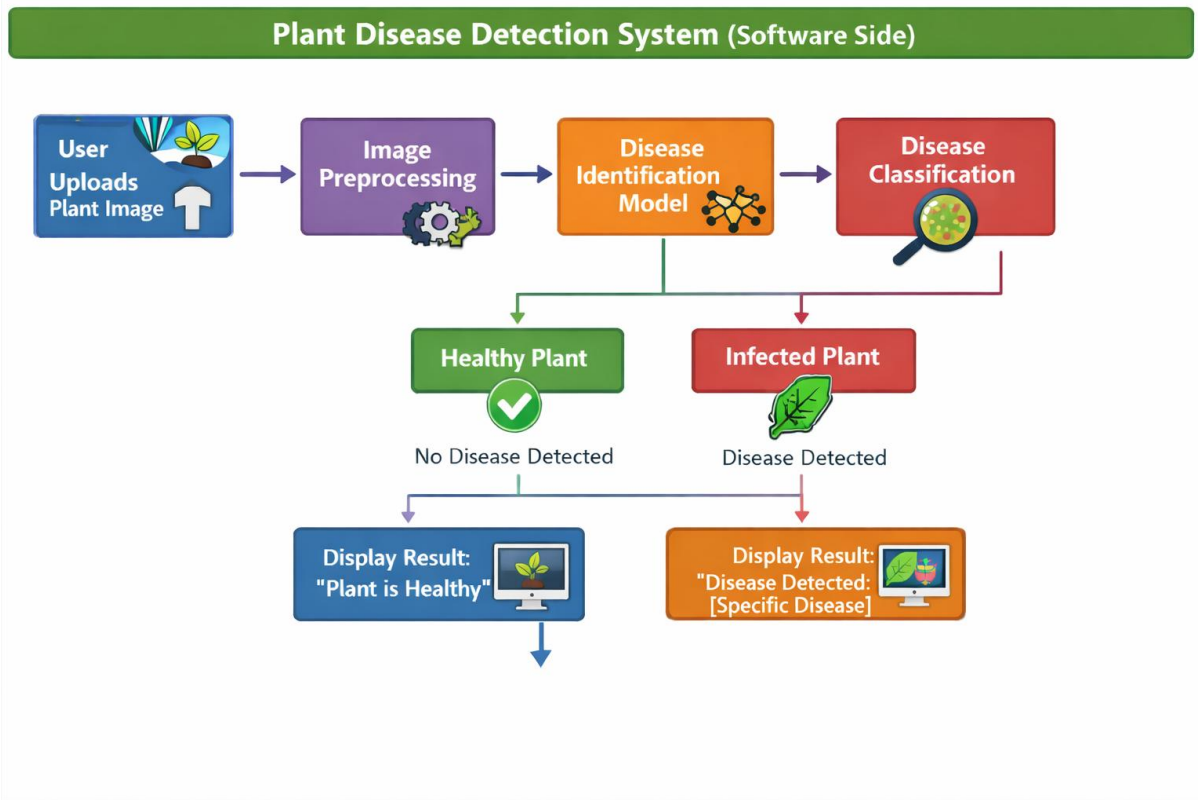


Fig 3.2 BLOCK DIAGRAM

3.3 DESCRIPTION OF COMPONENTS

3.3.1 ESP32 Microcontroller

The ESP32 microcontroller is the main controller of the system. It receives data from various sensors, processes the readings, and controls the operation of other components such as the relay module, motor pump, and LCD display. The ESP32 is widely used in IoT applications because of its high processing capability, built-in Wi-Fi support, and multiple input/output pins.

3.3.2 Soil Moisture Sensor

The soil moisture sensor is used to measure the moisture level present in the soil. It helps determine whether the soil is dry or contains sufficient water for plant growth. The sensor sends signals to the ESP32, which analyzes the data and decides whether irrigation is required for the plant.

3.3.3 MQ-135 Sensor

The MQ-135 sensor is used to detect harmful gases and air quality around the plant environment. It can sense gases such as ammonia, benzene, smoke, and carbon dioxide. Monitoring these gases helps identify environmental conditions that may affect plant health and growth.

3.3.4 Temperature Sensor

The temperature sensor is used to measure the surrounding temperature of the plant environment. Temperature plays an important role in plant growth and disease development. The sensor sends temperature readings to the ESP32, which helps in monitoring environmental conditions suitable for healthy plant growth.

3.3.5 Motor Pump

The motor pump is used to supply water to the plants when irrigation is required. When the soil moisture level becomes low, the ESP32 activates the motor pump through the relay module. This helps maintain proper soil moisture and ensures healthy plant growth.

3.3.6 Relay Module

The relay module acts as a switching device that allows the ESP32 to control high-power devices such as the motor pump. Since the ESP32 operates at low voltage, the relay module helps safely turn the pump ON or OFF based on the signals received from the microcontroller.

3.3.7 LCD Display

The LCD display is used to show important information such as soil moisture level, temperature, gas sensor readings, and system status. It allows users to easily monitor the plant condition and system operation in real time.

3.3.8 Buck Converter

The buck converter is used to regulate and step down the input voltage to the required level for the electronic components. It ensures that components like the ESP32 and sensors receive stable and safe voltage for proper operation.

3.3.9 Power Supply

The power supply unit provides the required electrical energy to operate all the components in the system. It ensures stable voltage and current for the ESP32 microcontroller, sensors, LCD display, relay module, and motor pump, enabling the entire system to function properly.

3.4 WORKING PRINCIPLE

3.4.1 HARDWARE IMPLEMENTATION

The hardware implementation of the system consists of a microcontroller, sensors, LCD display, relay module, pump, and power supply. All the components are connected to form a plant monitoring and automatic fertilizer supply system. When the system is powered on, the microcontroller initializes all the connected hardware components. The temperature and humidity sensor measures the surrounding environmental conditions of the plant. The soil moisture sensor is inserted into the soil to measure the moisture level present in

the soil. The gas sensor is used to detect volatile gases released by the plant, which may indicate plant stress or disease. The sensor outputs are given as input to the microcontroller. The microcontroller continuously reads the sensor values and processes the data. The measured parameters such as temperature, humidity, soil moisture level, and gas concentration are displayed on the LCD display for real-time monitoring. A threshold value of 200 is set for the gas sensor. When the detected gas value exceeds this threshold level, it indicates that the plant may be affected by disease because plants release volatile gases when they are under stress or infection.

When this condition occurs, the microcontroller sends a signal to the relay module. The relay acts as a switching device that controls the pump. Once the relay is activated, the pump turns on and supplies fertilizer to the plant automatically. The entire hardware system continuously monitors the plant conditions and responds accordingly. This implementation helps in early detection of plant stress and provides automatic fertilizer supply to maintain plant health.

3.4.2 SOFTWARE IMPLEMENTATION

The software implementation of the system is developed as a web-based platform that allows farmers to monitor and analyze plant health easily. A user interface is created where users can access the system through a signup and login page to ensure secure access. After logging in, the user can view basic analysis information and access the image upload section available on the dashboard. In this section, the farmer uploads an image of the plant leaf using the upload box provided in the webpage.

Once the image is uploaded, the system performs image preprocessing such as resizing and noise removal to improve the quality of the input image. The

processed image is then analyzed using an image processing and disease prediction model. The model examines the visual features of the leaf such as color changes, spots, or damaged areas to identify possible diseases. Based on this analysis, the system predicts whether the plant is healthy or affected by a disease and displays the detected disease name on the webpage.

The software mainly focuses on detecting diseases in three crops: rice, sugarcane, and banana. The web application is designed to be simple and user-friendly so that farmers can easily upload images and view results without technical knowledge. The system stores user details and uploaded images for future reference and analysis. Since the platform is web-based, farmers can access it at any time using a computer or mobile device with an internet connection, which makes plant disease detection faster and more convenient.

3.5 FLOWCHART OF OPERATION

3.5.1 Hardware Flowchart

1. Short Step-by-Step Description of the Flowchart
2. Start – The system begins operation.
3. Initialize System – ESP32 and all sensors are powered ON and initialized.
4. Read Sensor Values – Temperature, Humidity, Soil Moisture, and Gas sensor data are collected.
5. Display Data – The sensor values (T, H, S, G) are displayed on the LCD.
6. Check Gas Level – The system checks whether the gas level is greater than 200.
7. If Gas > 200 – Plant disease is suspected. The pump turns ON and fertilizer is sprayed automatically.
8. Turn OFF Pump – After a few seconds, the pump is turned OFF.

9. If Gas ≤ 200 – The system continues normal monitoring.

10. Repeat Process – The system continuously monitors the sensors again.

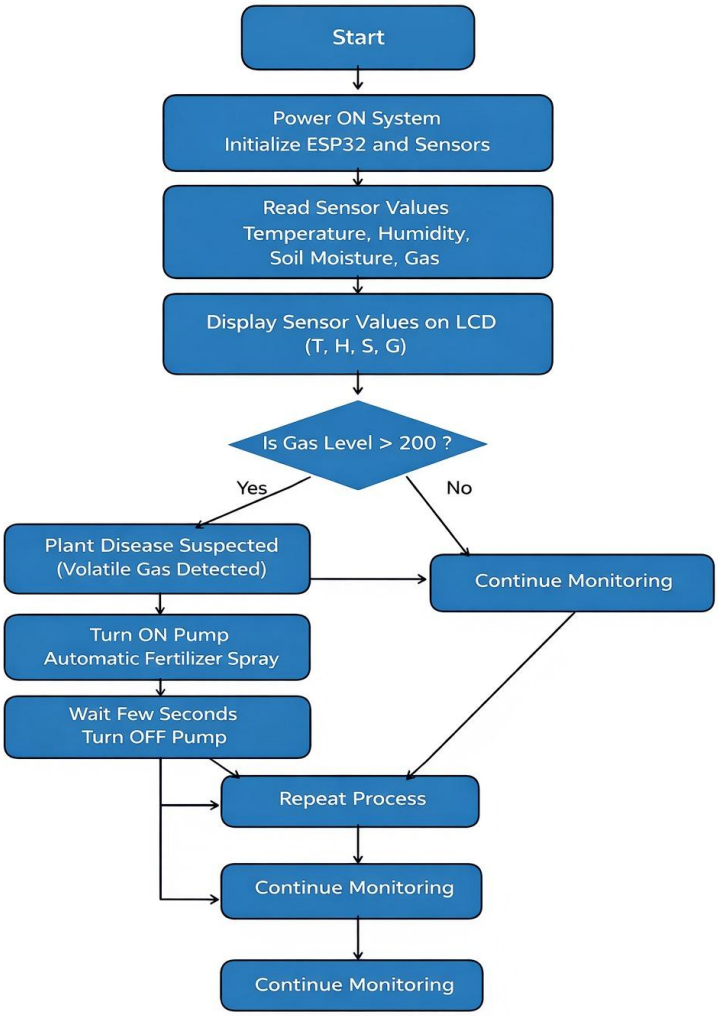


Fig 3.3 HARDWARE FLOWCHART

3.5.2 Software Flowchart

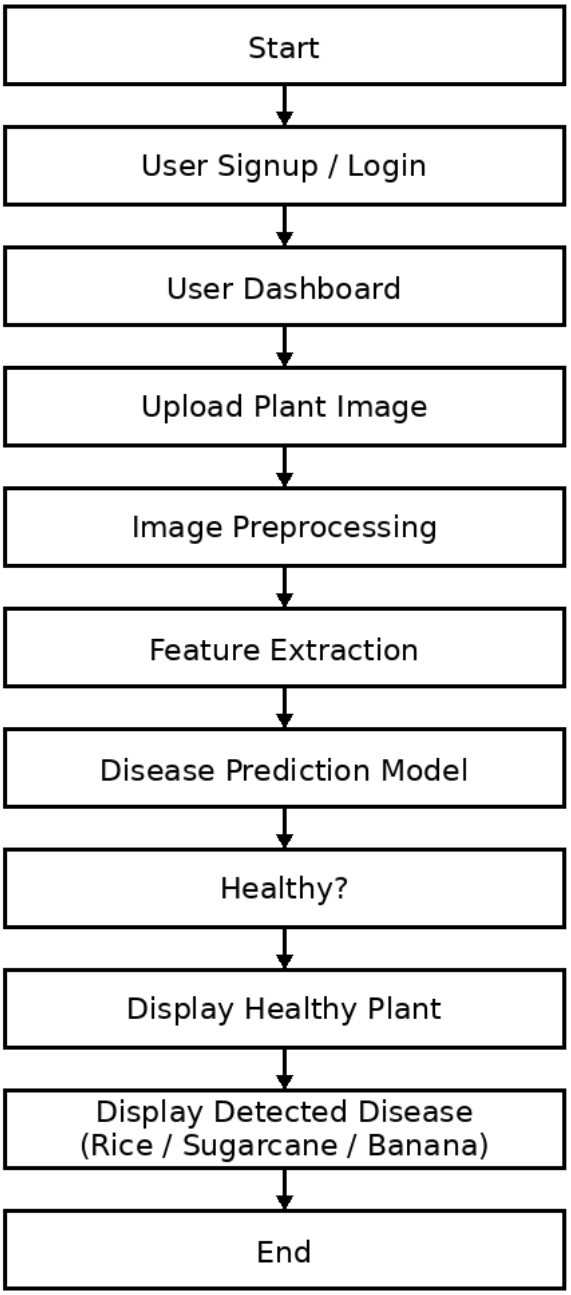


Fig 3.4 SOFTWARE FLOWCHART

3.6 ADVANTAGES OF THE PROPOSED SYSTEM

1. Real-Time Monitoring: The system continuously monitors important plant parameters such as temperature, humidity, soil moisture, and volatile gas levels using sensors.
2. Early Disease Detection: The gas sensor helps detect volatile gases released by plants when they are under stress or affected by disease, enabling early identification of plant health issues.
3. Automatic Fertilizer Supply: When the gas level exceeds the predefined threshold, the system automatically activates the pump to supply fertilizer to support plant recovery.
4. Web-Based Disease Detection: Farmers can upload plant leaf images through a web interface, and the system analyzes the image to identify whether the plant is healthy or affected by a disease.
5. User-Friendly Interface: The web platform includes features such as signup, login, and an image upload section, making it easy for farmers to use the system.
6. Cost-Effective System: The system uses affordable components such as sensors, ESP32, and other modules, making it suitable for practical agricultural applications.
7. Focus on Major Crops: The system is designed to detect diseases in important crops such as rice, sugarcane, and banana, which are widely cultivated.
8. Improved Plant Health Monitoring: Continuous monitoring of environmental conditions and disease detection helps farmers take timely action to maintain healthy crop growth.

CHAPTER 4

RESULTS AND DISCUSSION

4.1. HARDWARE SETUP OF THE PLANT MONITORING SYSTEM

This figure shows the complete hardware setup of the proposed plant monitoring system. The system consists of a microcontroller board, temperature and humidity sensor, gas sensor, soil moisture sensor, relay module, water pump, battery power supply, and LCD display. All components are connected using jumper wires and mounted on a base board. The sensors are placed in positions suitable for measuring plant environmental conditions. The system is designed to monitor plant parameters and automatically respond when abnormal conditions are detected.

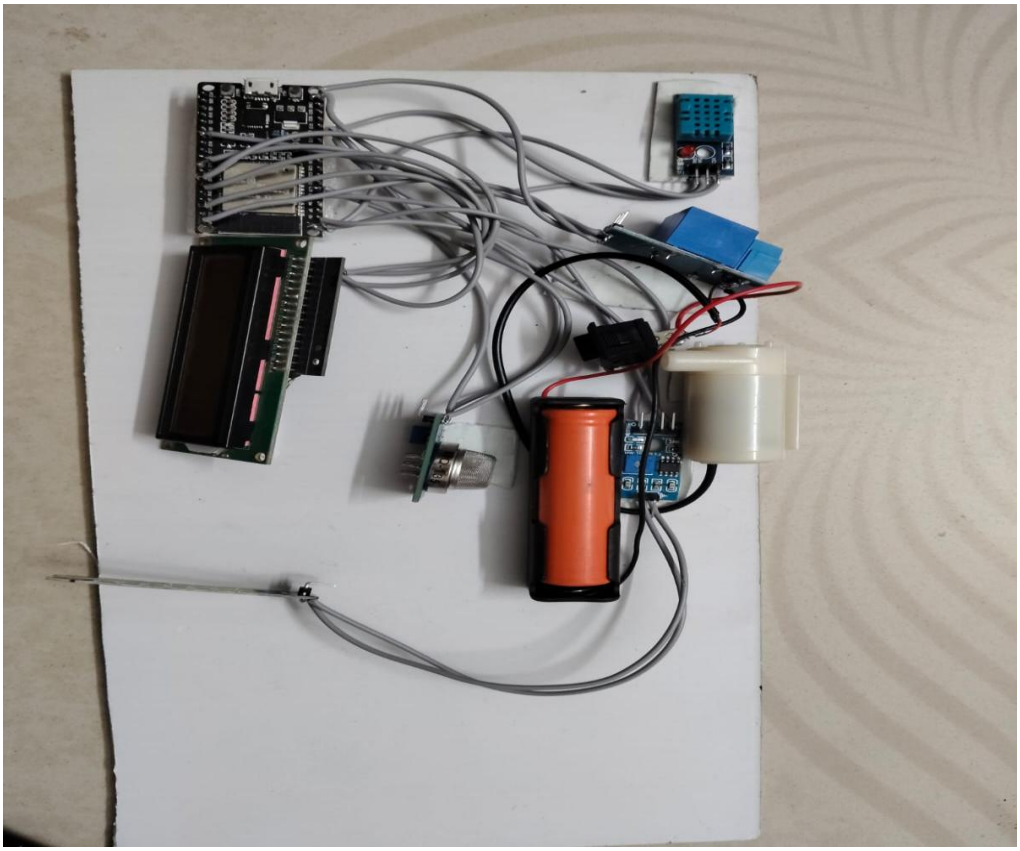


Fig 4.1 Hardware Setup

4.2 SYSTEM OUTPUT DISPLAY UNDER NORMAL CONDITION

This figure shows the working condition of the system when the sensor values are within the normal range. The LCD display shows the measured parameters such as temperature (T), humidity (H), soil moisture level (S), and gas level (G). In this condition, the gas value is below the threshold level of 200, which indicates that the plant is in a healthy condition. Therefore, the relay and pump remain in the OFF state while the system continues monitoring the plant environment.

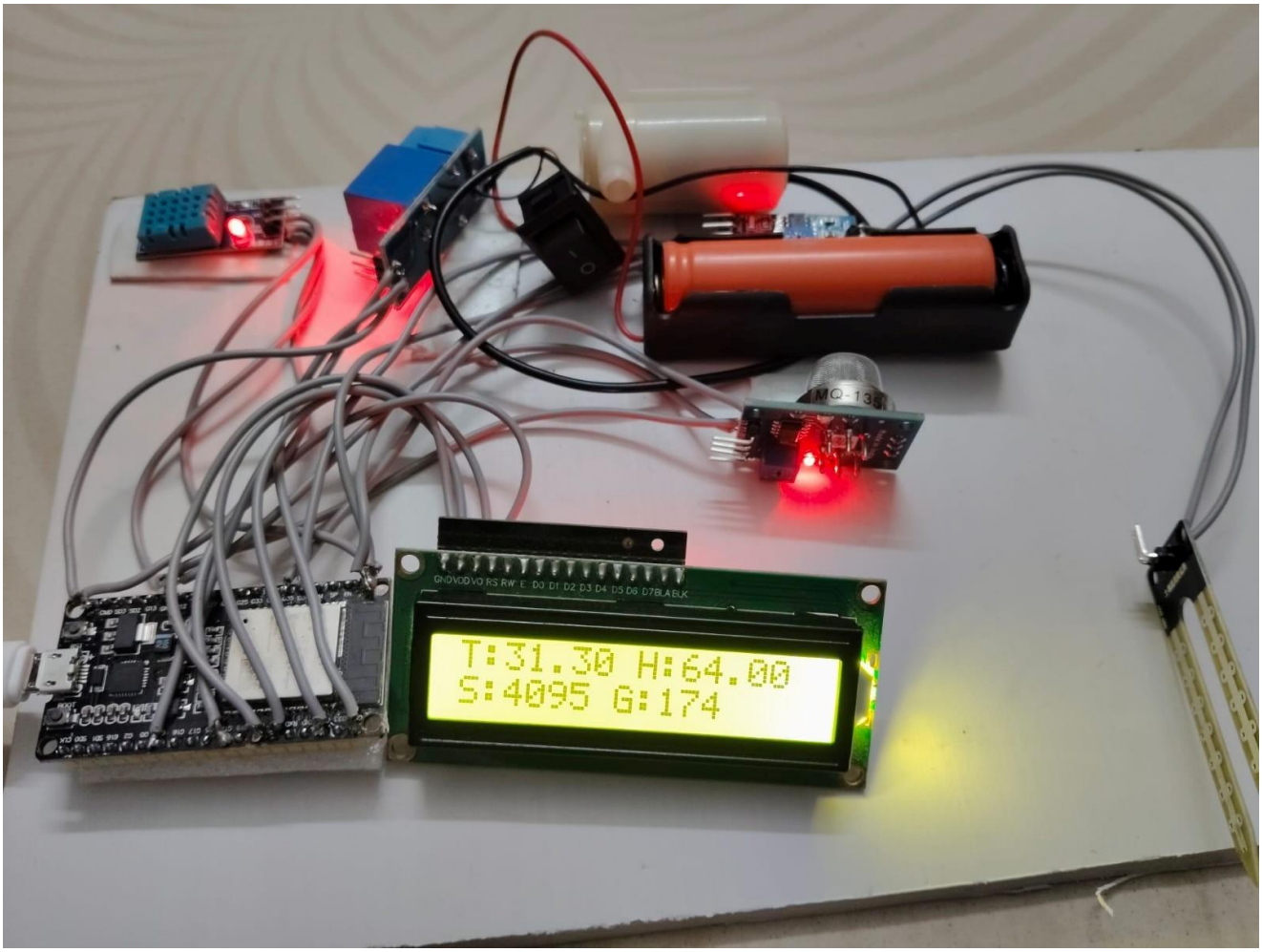


Fig 4.2 Before Volatile Gas Released

4.3 SYSTEM RESPONSE WHEN GAS LEVEL EXCEEDS THRESHOLD

This figure shows the system response when the gas sensor detects a value higher than the threshold level. The LCD display indicates the updated sensor values where the gas concentration exceeds the preset limit. This condition suggests that the plant may be under stress or affected by disease. As a result, the relay module is activated and the pump turns ON to supply fertilizer automatically. This demonstrates the automatic response capability of the proposed system.

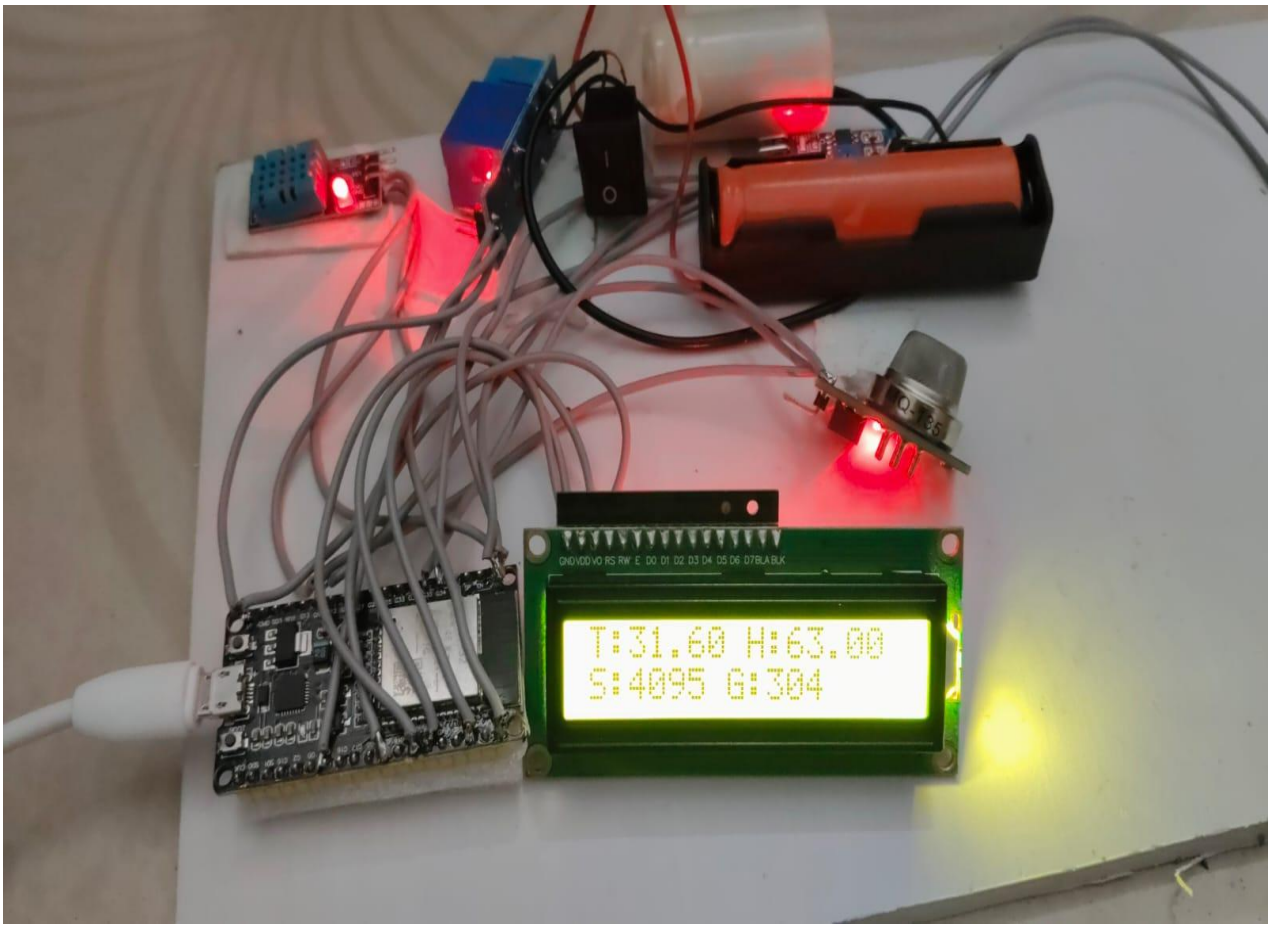


Fig 4.3 After Volatile Gas Released

4.4 SOFTWARE RESULTS

4.4.1 Home page

The home page introduces the LeafCare AI platform, which helps farmers detect plant diseases using artificial intelligence. It provides navigation options such as Home, About, Login, and Signup, allowing users to access the system easily. The page highlights key features like fast disease detection, neural analysis, and data-driven insights to support precision agriculture.

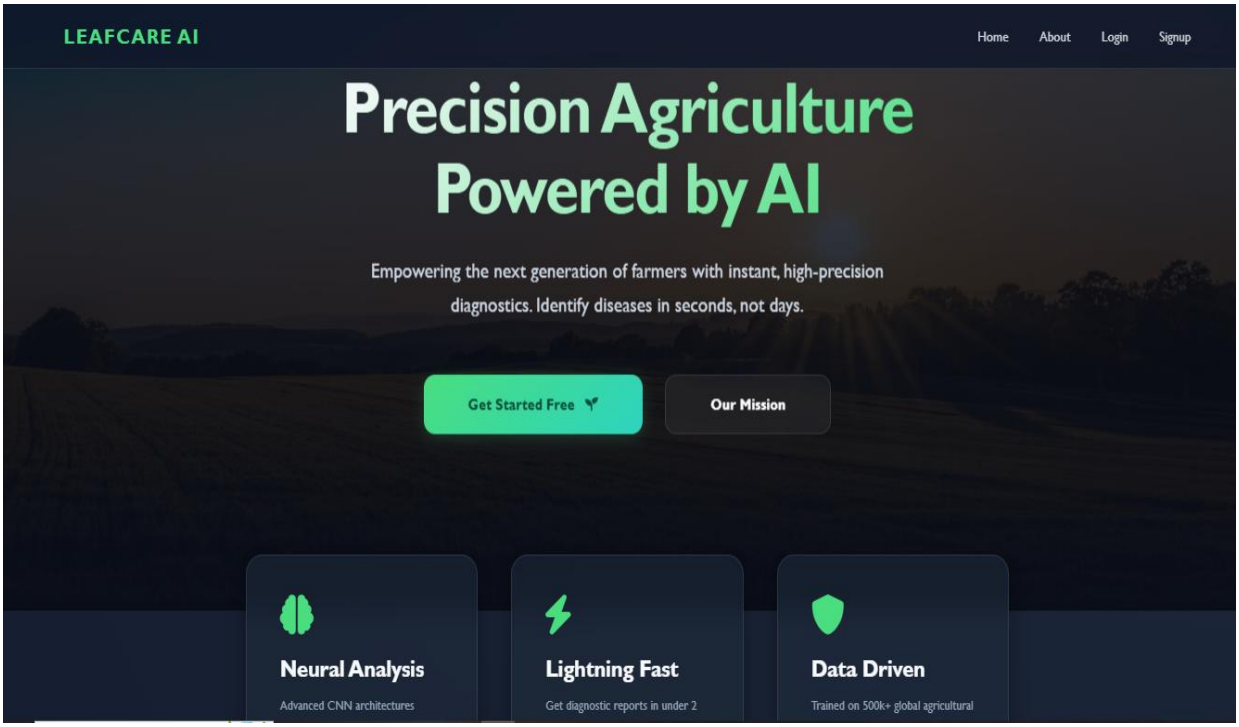
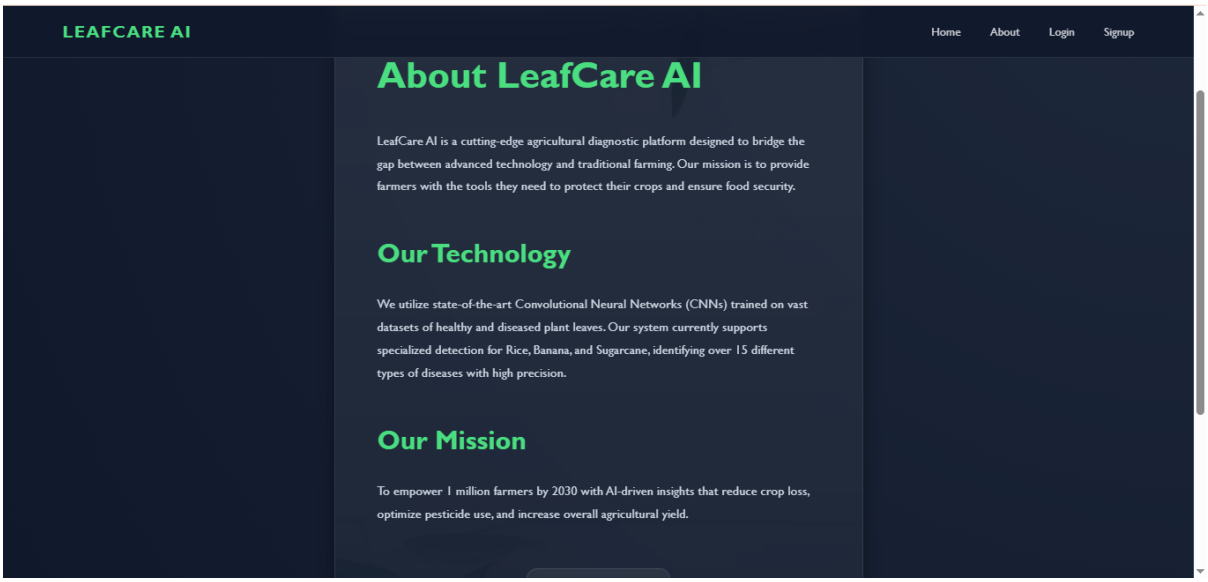


Fig 4.4 Home page

4.4.2 About page

The About page provides information about the LeafCare AI platform and its purpose. It explains how the system uses artificial intelligence and convolutional neural networks to detect plant diseases from leaf images. The page also describes the technology used and the mission of the platform, which is to help farmers protect crops, reduce crop loss, and improve agricultural productivity.



4.4.3 SignUp page

The Signup page allows new users to create an account in the system. Users enter their username, email address, and password to register. After successful registration, they can log in to the platform and access features such as uploading plant images and viewing disease detection results.

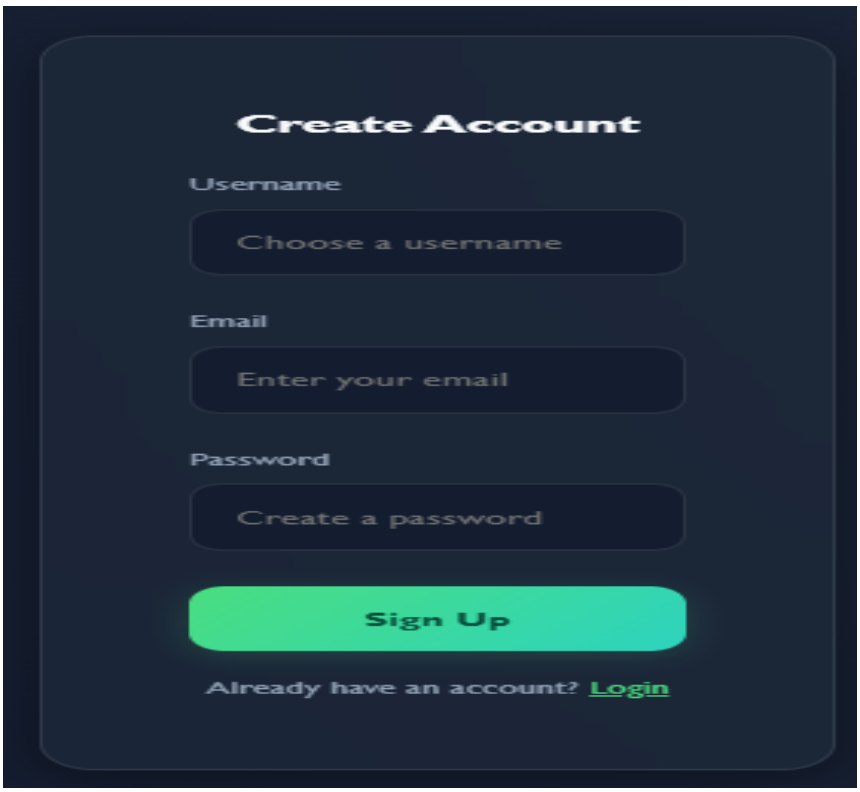


Fig 4.5 SignUp page

4.4.4 Login page

The Login page allows registered users to securely access the system by entering their email or username and password. After successful login, users can access the dashboard, upload plant images, and view plant disease detection results.

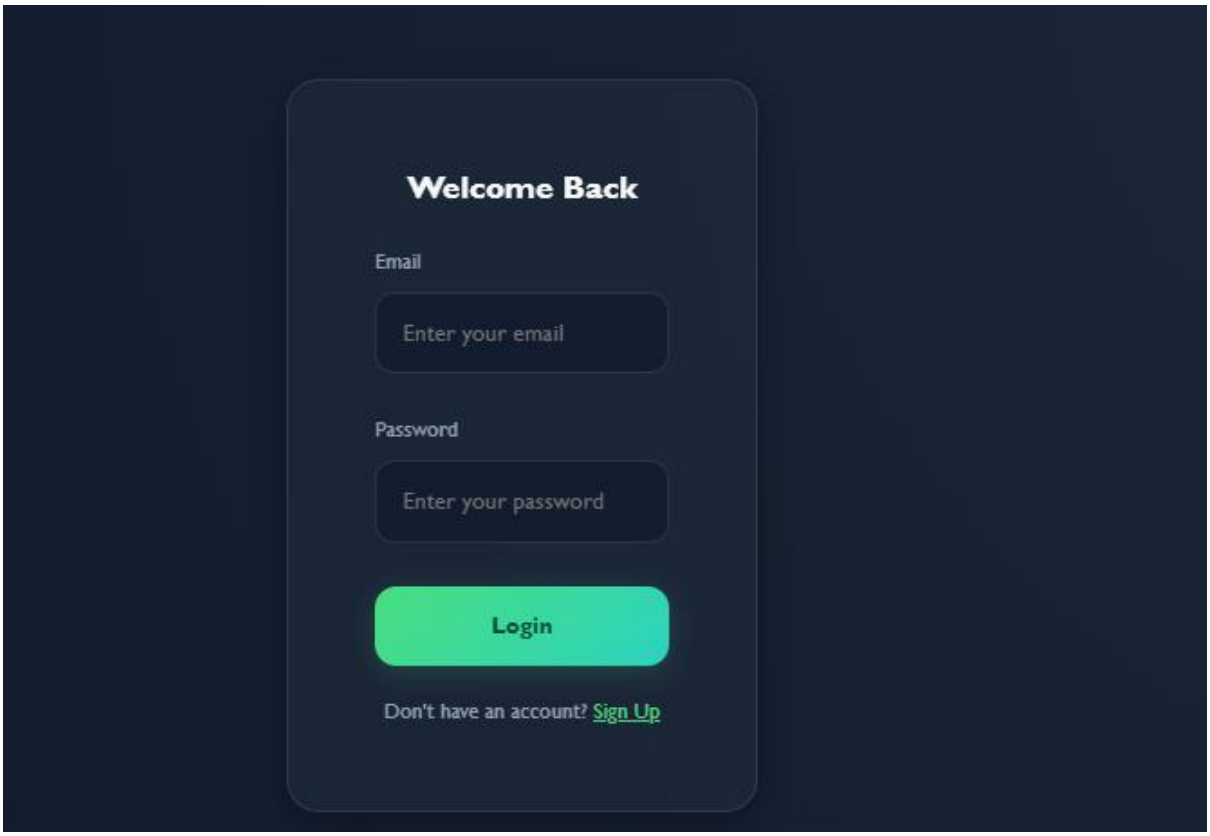


Fig 4.6 Login page

4.4.5 AI Diagnostics Page

The AI Diagnostics page allows users to upload an image of a plant leaf for disease detection. The system analyzes the uploaded image using an AI-based model and identifies whether the plant is healthy or affected by a disease. It also provides tips to help users upload clear images for better detection accuracy.

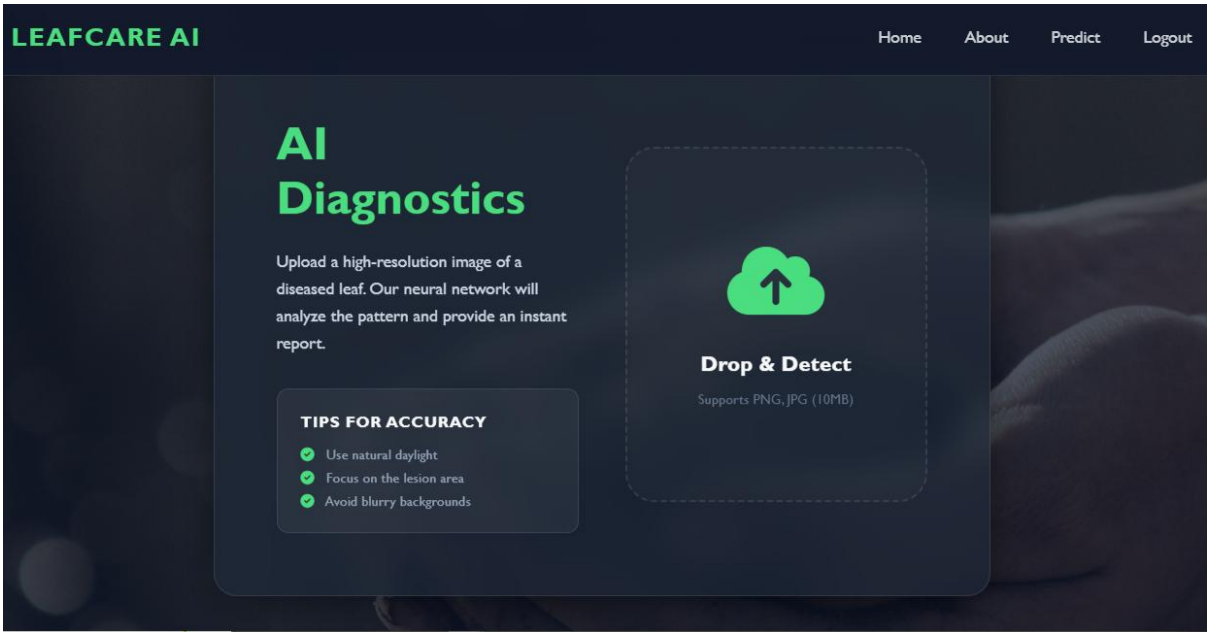


Fig 4.7 AI Diagnostics Page

4.4.6 Diagnostic Result Page

(i) Banana Leaf – Healthy Detection Result.

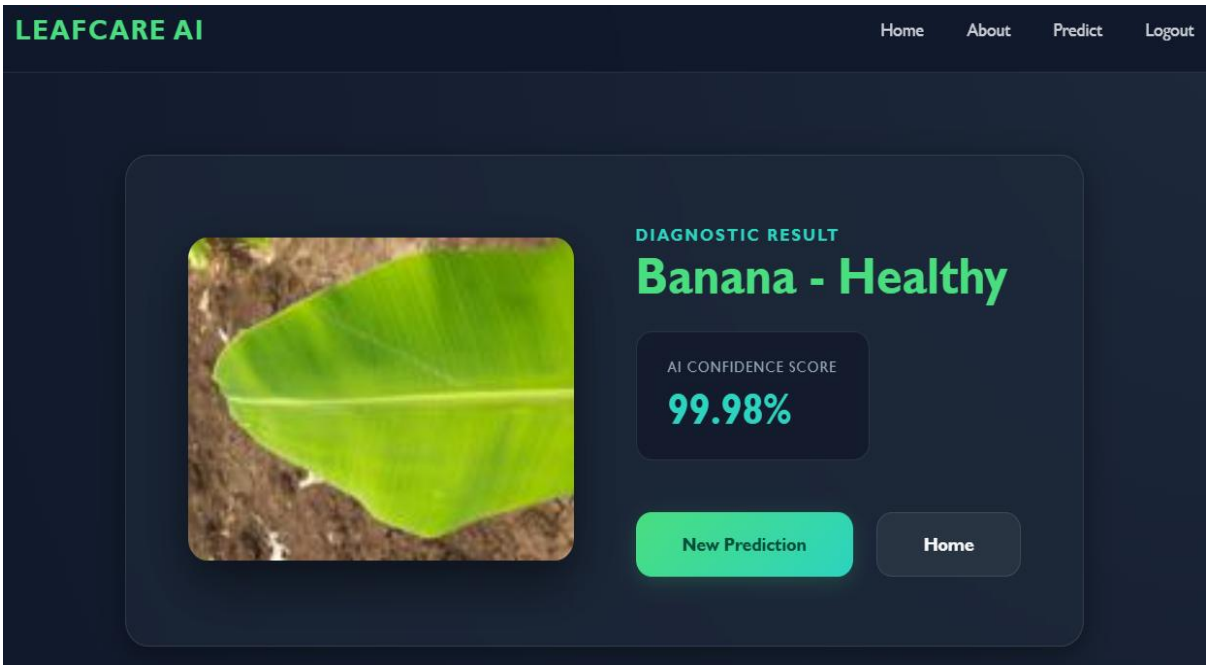


Fig 4.8 Diagnostic Result Page(i)

4.5 DISEASED PLANTS ANALYSE

4.5.1 Banana – Panama Disease

The system detects Banana – Panama Disease after analyzing the uploaded leaf image and provides an AI confidence score of 100%. The result helps users quickly identify the plant disease and take necessary actions. Options are also provided to perform a new prediction or return to the home page.

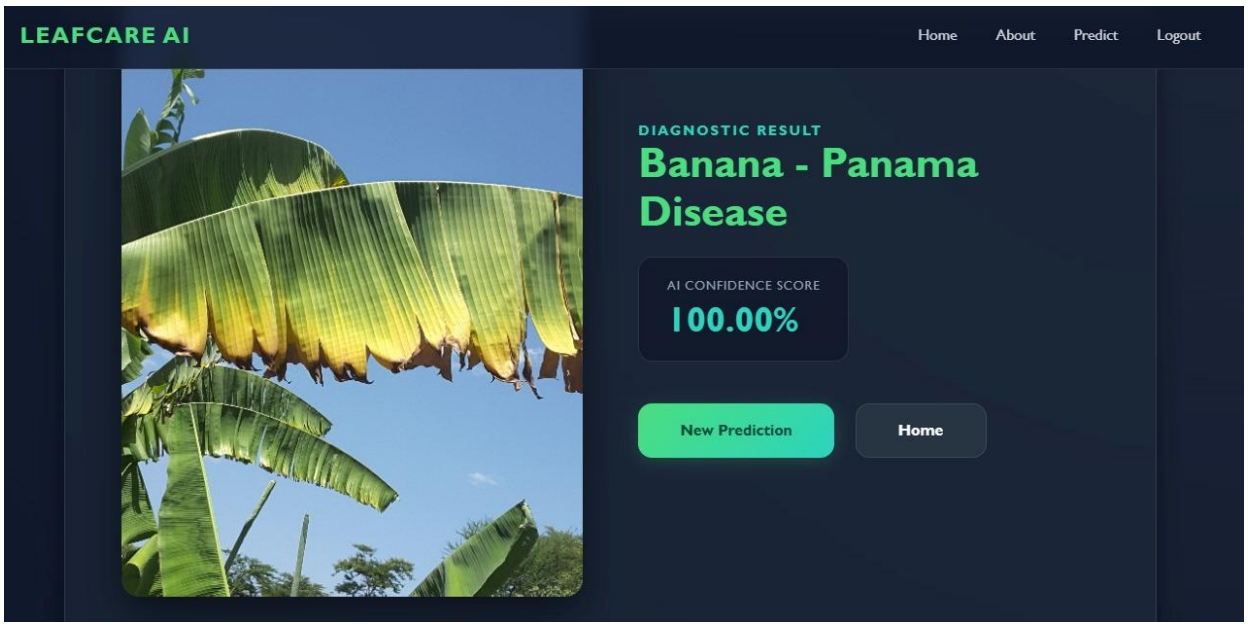


Fig 4.9 Banana – Panama Disease

4.5.2 Rice – Brown Spot

The system processes the uploaded rice leaf image and identifies the presence of Brown Spot disease. The prediction is generated using the trained model, providing a confidence score of 100%. This result enables users to understand the plant condition and supports timely decision-making for proper crop management.

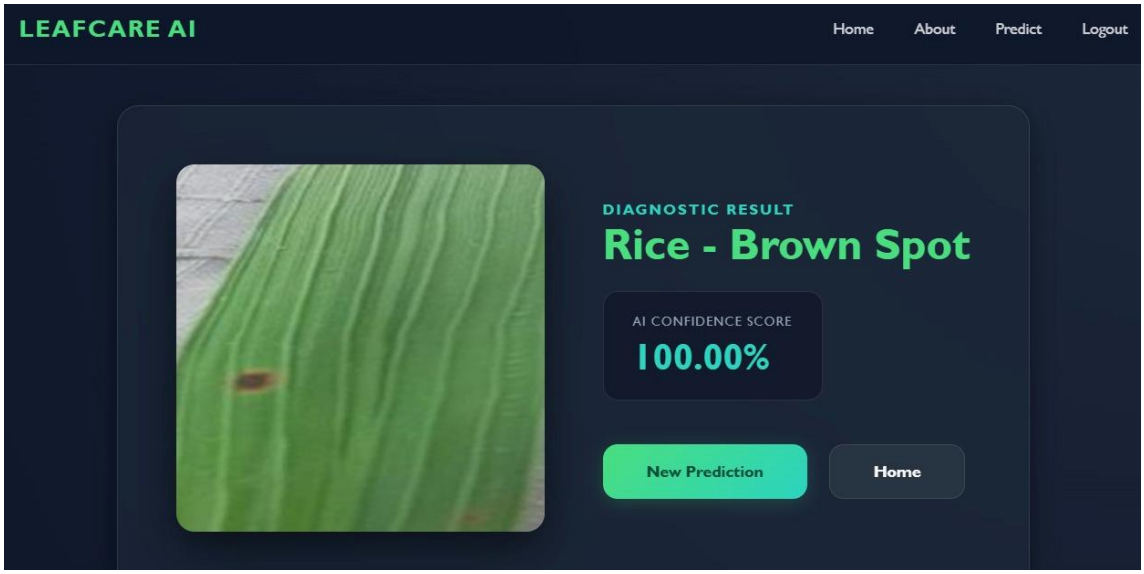


Fig 4.10 Rice – Brown Spot

4.5.3 SugarCane - RedRot

The uploaded sugarcane leaf image is analyzed by the model, which detects Red Rot disease with a confidence score of 99.99%. This prediction helps users quickly recognize the disease and take suitable actions to maintain crop health.

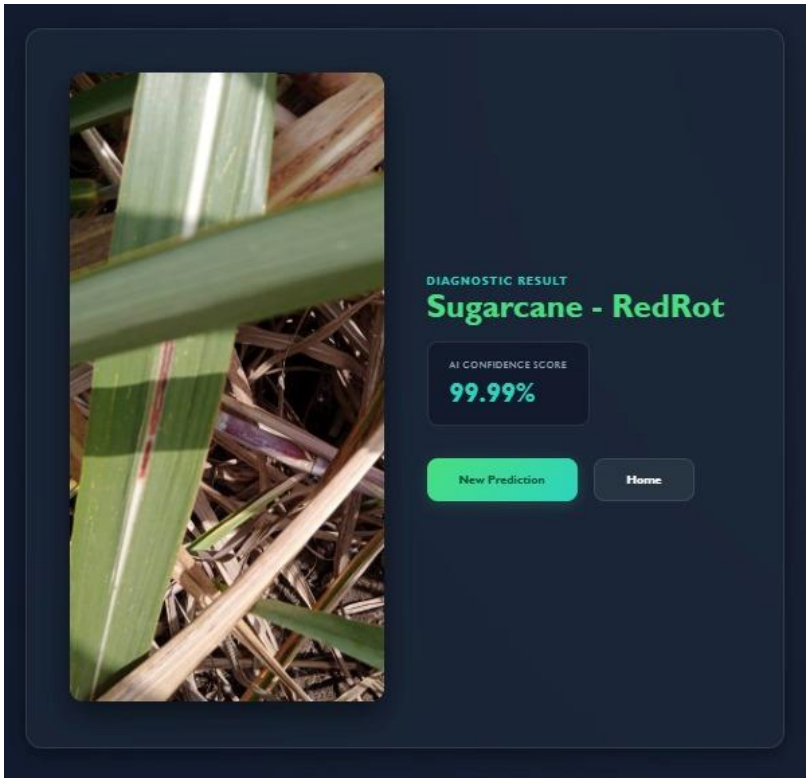


Fig 4.11 SugarCane - RedRot

4.6 DISCUSSION OF OBSERVATIONS

The developed system was tested to monitor plant conditions and detect possible diseases using both hardware sensors and image-based analysis. The hardware section successfully measured parameters such as temperature, humidity, soil moisture, and volatile gas levels. It was observed that when the gas sensor value exceeded the predefined threshold of 200, the system identified it as a possible sign of plant stress or disease and automatically activated the fertilizer pump.

In the software section, the web-based platform allowed users to upload plant leaf images for disease detection. The AI model analyzed the images and correctly identified whether the plant was healthy or affected by a disease in rice, sugarcane, or banana plants. The results were displayed with a confidence score, showing the reliability of the prediction.

Overall, the system demonstrated that combining sensor-based monitoring with AI-based image analysis can help farmers detect plant health issues quickly and take timely action to protect crops.

4.7 ADVANTAGES AND LIMITATIONS

4.7.1 Advantages

- The system helps in early detection of plant diseases using both sensors and image analysis.
- Farmers can easily upload plant images through the web interface to check plant health.
- Automatic fertilizer pumping is activated when the gas level exceeds the threshold value.
- The system continuously monitors environmental parameters such as temperature, humidity, soil moisture, and gas levels.

- The web-based platform can be accessed anytime and from anywhere.
- It supports important crops such as rice, sugarcane, and banana.

4.7.2 Limitations

- The accuracy of detection depends on the quality of the uploaded leaf image.
- The system currently supports only a limited number of plant types.
- An internet connection is required to use the web application.
- Sensor readings may vary due to environmental conditions or improper sensor placement.
- Regular maintenance and calibration of sensors may be required for accurate results.

CHAPTER 5

BUDGET ESTIMATION

5.1 COMPONENT LIST

The following components were used in the implementation of the LoRa-Based Smart Community Water Quality Monitoring and Early Warning System:

S.No.	Component Name	Specification	Quantity
1	ESP32 Microcontroller	Dual Core Wi-Fi + Bluetooth MCU	1
2	Soil Moisture sensor	Analog Soil Moisture Detection Sensor	1
3	MQ-135 Gas Sensor	Air Quality / Harmful Gas Detection Sensor	1
4	Temperature Sensor	Temperature Monitoring Sensor (e.g., LM35 / DHT11)	1
5	LCD Display	16×2 Alphanumeric Display	1
6	Relay Module	Used to Control Motor Pump	1
7	Fertilizer Pump	Mini Pump for Automatic Fertilizer Supply	1
8	Buck Converter	DC-DC Step-Down Voltage Converter	1
9	Power Supply	5V / 12V Adapter or Rechargeable Battery	1

5.2 COST ESTIMATION TABLE

Component	Unit Price (₹)	Quantity	Total (₹)
ESP32 Microcontroller	390	1	390
Soil Moisture sensor	60	1	60

Component	Unit Price (₹)	Quantity	Total (₹)
MQ-135 Gas Sensor	125	1	125
Temperature Sensor	60	1	60
LCD Display	200	1	200
Relay Module	60	1	60
Fertilizer Pump	60	1	60
Buck Converter	60	1	60
Power Supply/Battery	110	1	110

5.3 TOTAL BUDGET

The total estimated cost for developing the plant health monitoring and disease detection system is approximately ₹1125, including the required hardware components and basic assembly materials. The system uses affordable components such as sensors, an ESP32 microcontroller, and other supporting modules, making it suitable for academic projects and small-scale agricultural applications. The overall cost is kept low by selecting easily available and cost-effective components. This makes the system economical and practical for student projects while still providing important features such as plant condition monitoring and early disease detection.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The proposed system provides an efficient solution for monitoring plant health and detecting possible diseases using both hardware and software technologies. In the hardware section, sensors are used to continuously monitor important environmental parameters such as temperature, humidity, soil moisture, and volatile gas levels around the plant. The collected data helps in understanding the plant's growing conditions. When the gas sensor detects a value that exceeds the predefined threshold, it indicates that the plant may be under stress or affected by a disease. In such cases, the system automatically activates the fertilizer pump to support plant growth and recovery.In the software section, a web-based platform is developed to allow farmers to easily access the system. The platform includes features such as signup, login, and an image upload interface. Farmers can upload images of plant leaves, and the system analyzes the images using an AI-based disease detection model. The system then predicts whether the plant is healthy or affected by a disease and displays the result to the user. The project mainly focuses on detecting diseases in three important crops: rice, sugarcane, and banana. Overall, the system helps farmers monitor plant conditions and detect diseases at an early stage. By combining sensor-based monitoring with image-based disease detection, the system provides a simple and cost-effective solution for improving plant health management. This approach can help reduce crop loss, support better decision-making for farmers, and contribute to improved agricultural productivity in the future.

6.2 FUTURE SCOPE

The proposed system provides an effective solution for monitoring plant health and detecting possible diseases using both hardware sensors and software-based analysis. The hardware section continuously monitors environmental parameters such as temperature, humidity, soil moisture, and volatile gas levels. When the gas level exceeds the predefined threshold, it indicates possible plant stress or disease, and the system automatically activates the fertilizer pump. The software section includes a web-based platform where farmers can sign up, log in, and upload plant leaf images for disease detection. The system analyzes the images using an AI-based model and identifies whether the plant is healthy or affected by a disease. The project mainly focuses on three crops: rice, sugarcane, and banana. Overall, the system helps farmers monitor plant conditions and detect diseases at an early stage, which can help reduce crop loss and improve agricultural productivity.

SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed to blue print to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030, In 2015, 195 nations agreed as a blue print that they can change the world for the better. The project is based on one of the 17 goals.

Questions	Answers
Which SDGs does the project directly address?	SDG 2 – Zero Hunger and SDG 12 – Responsible Consumption and Production.
What strategies or actions are being implemented to achieve these goals?	By developing an automated plant disease detection system that analyzes crop images using machine learning to identify diseases.
How is progress measured and reported in relation to the SDGs?	Through testing model accuracy, analyzing disease detection results, and evaluating system performance using different plant leaf image samples.
How were these goals identified as relevant to the project's objectives?	The project addresses crop disease detection and supports sustainable agriculture and improved food production.
Are there any partnerships or collaborations in place to enhance this impact?	Developed with guidance from faculty mentors and student team collaboration.

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